MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NOR, DIGITAL, KEY CODE EMP

This amendment forms a part of Military Specification MIL-M-28787/263(EC), dated 3 February 1982, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, AND 6

TABLE I: Delete and substitute new table I as printed on pages 2, 3, and 4 of this amendment.

PAGE 7

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

PAGE 8

6.4, Output drive current (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

PAGE 17

FIGURE 6: Delete and substitute new figure 6 as printed on page 5 of this amendment.

PAGE 20

FIGURE 7: Delete and substitute new figure 7 as printed on page 6 of this amendment.

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FIGURE 8: Delete and substitute new figure 8 as printed on page 7 of this amendment.

PAGE 25

FIGURE 9: Delete and substitute new figure 9 as printed on page 8 of this amendment.

PAGE 28

FIGURE 10: Delete and substitute new figure 10 as printed on page 9 of this amendment.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N502)

FSC 5963

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage (negative)	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage (positive)	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Output voltage (high level), I_{OH} = 0	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E12	Output voltage (low level), I_{OL} = 0	VII	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F12	Output voltage (high level), V_{DD} = 5 V	VIII	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
G1-G24	Output voltage (low level), V_{DD} = 5 V	IX	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
H1-H12	Output voltage (high level), V_{DD} = 15 V	X	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
J1-J12	Output voltage, (low level), V_{DD} = 15 V	XI	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
K1-K24	Input current (high level)	XII	X	0	200	0	200	0	200	0	200	0	200	nA
L1-L24	Input current (low level)	XIII	X	0	-200	0	-200	0	-200	0	-200	0	-200	nA
M1	Power supply current, low level	XIV	X	0	6	0	6	0	60	0	6	0	98	µA
M2	Power supply current, high level	XIV	X	0	6	0	6	0	60	0	6	0	98	µA
N1-N12	Transition time, (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XV	S	20	200	20	200	22	220	15	150	23	236	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
N13-N24	Transition time, (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XV	Q	68	680	68	680	75	750	51	515	80	803	ns
P1-P12	Transition time, (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	20	200	20	200	22	220	15	150	23	236	ns
P13-P24	Transition time(t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	68	680	68	680	75	750	51	515	80	803	ns
Q1-Q24	Propagation delay time, (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XVII	S	25	250	25	250	27	275	19	190	29	294	ns
Q25-Q48	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	45	450	45	450	50	500	34	340	53	531	ns
R1-R24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	XVIII	S	25	250	25	250	27	275	19	190	29	294	ns
R25-R48	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XVIII	Q	45	450	45	450	50	500	34	340	53	531	ns
S1-S12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	10	100	10	100	11	110	7	75	12	119	ns
T1-T12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XX	Q	10	100	10	100	11	110	7	75	12	119	ns
U1-U24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XXI	Q	10	100	10	100	11	110	7	75	12	119	ns
V1-V24	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XXII	Q	10	100	10	100	11	110	7	75	12	119	ns
W1-W12	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XXIII	Q	8	80	8	80	9	90	6	60	10	96	ns
X1-X12	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XXIV	Q	8	80	8	80	9	90	6	60	10	96	ns

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Y1-Y24	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XXV	Q	8	80	8	80	9	90	6	60	10	96	ns
Z1-Z24	Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXVI	Q	8	80	8	80	9	90	6	60	10	96	ns
AA	Power supply filter capacitance	XXVII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
BB1,BB2	C2 and C3 capacitance	XXVIII	X	176	264	176	264	176	264	176	264	176	264	pF
CC1-CC24	Input capacitance	XXIX	Q	2	15	2	15	2	15	2	15	2	15	pF

NOTES:

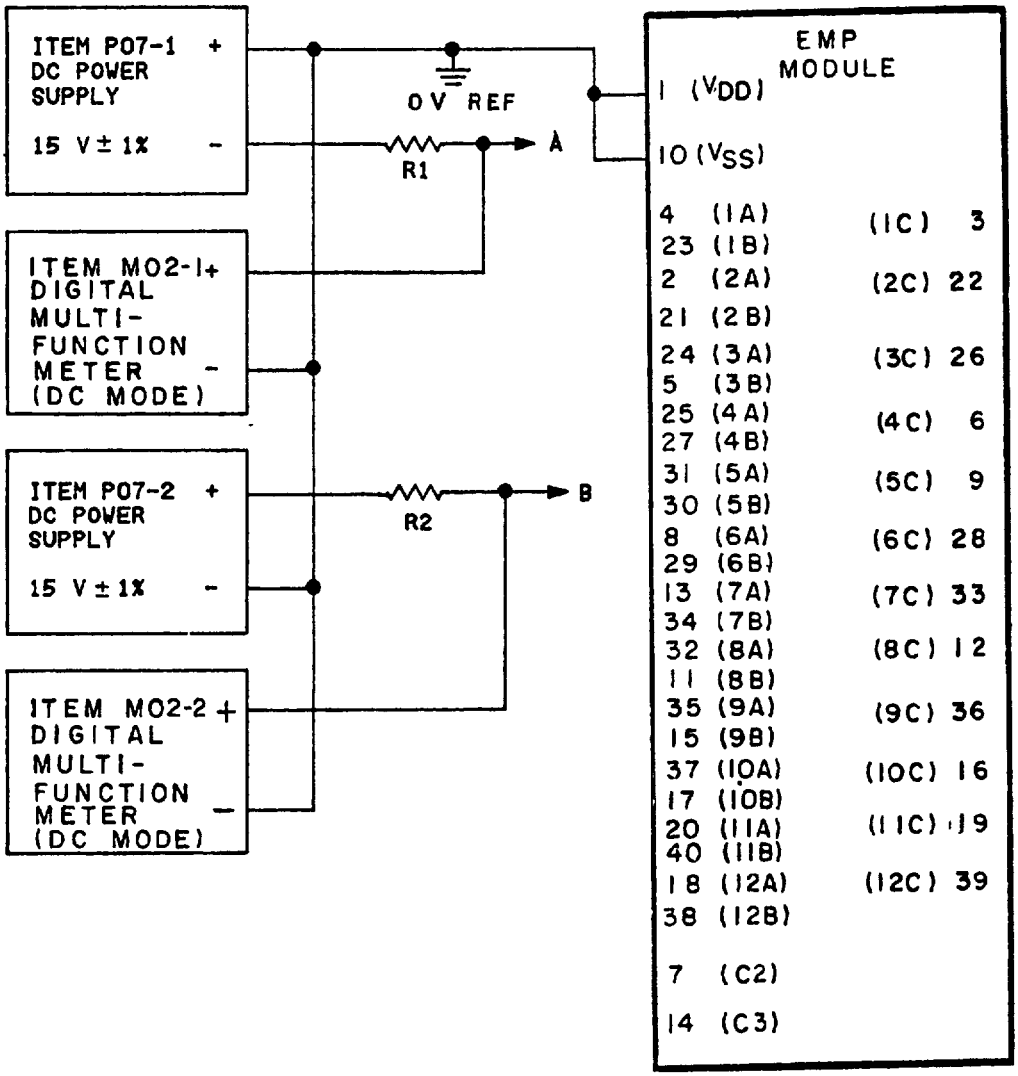
1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified, limits are for initial and end-of-life.

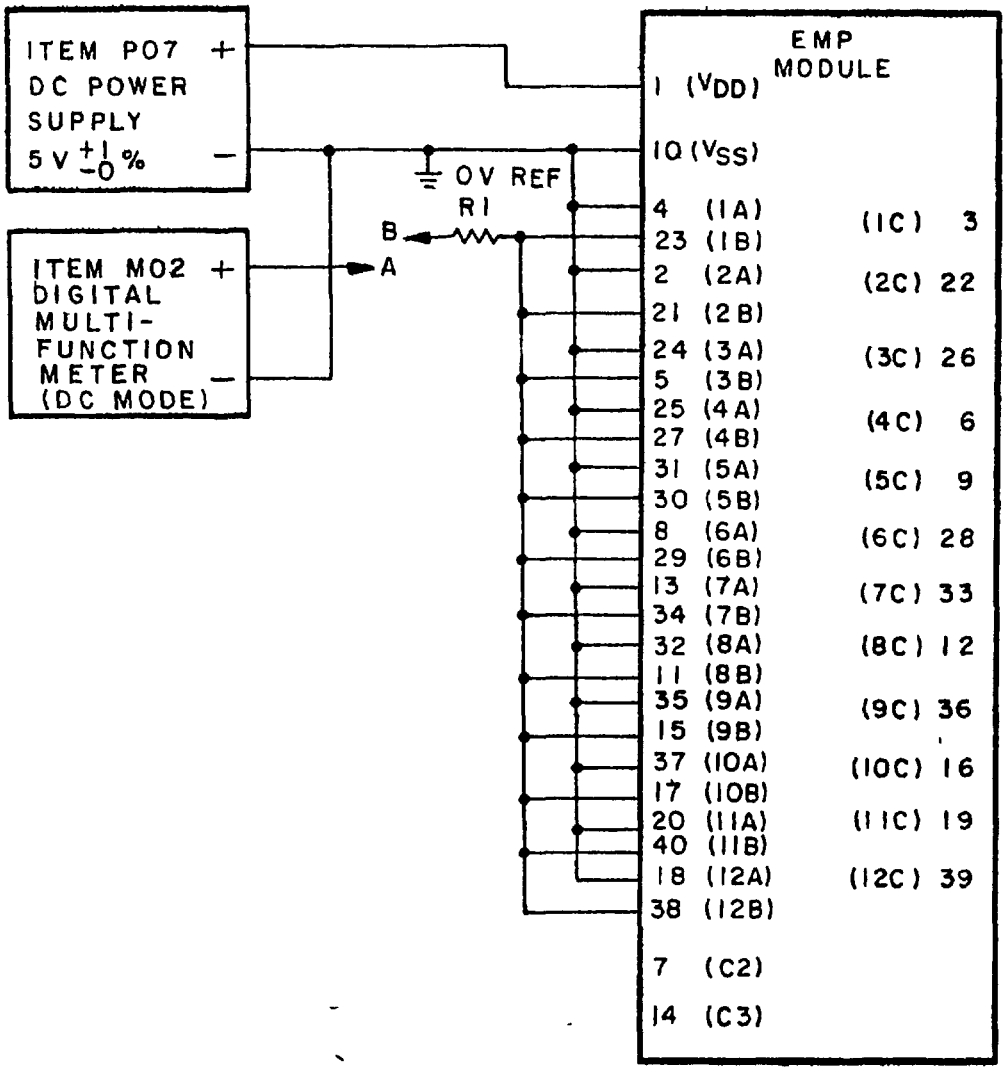
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NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since items P07-1 and P07-2 and items M02-1 and M02-2 are not used simultaneously, only one item P07 and one item M02 is required.

FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

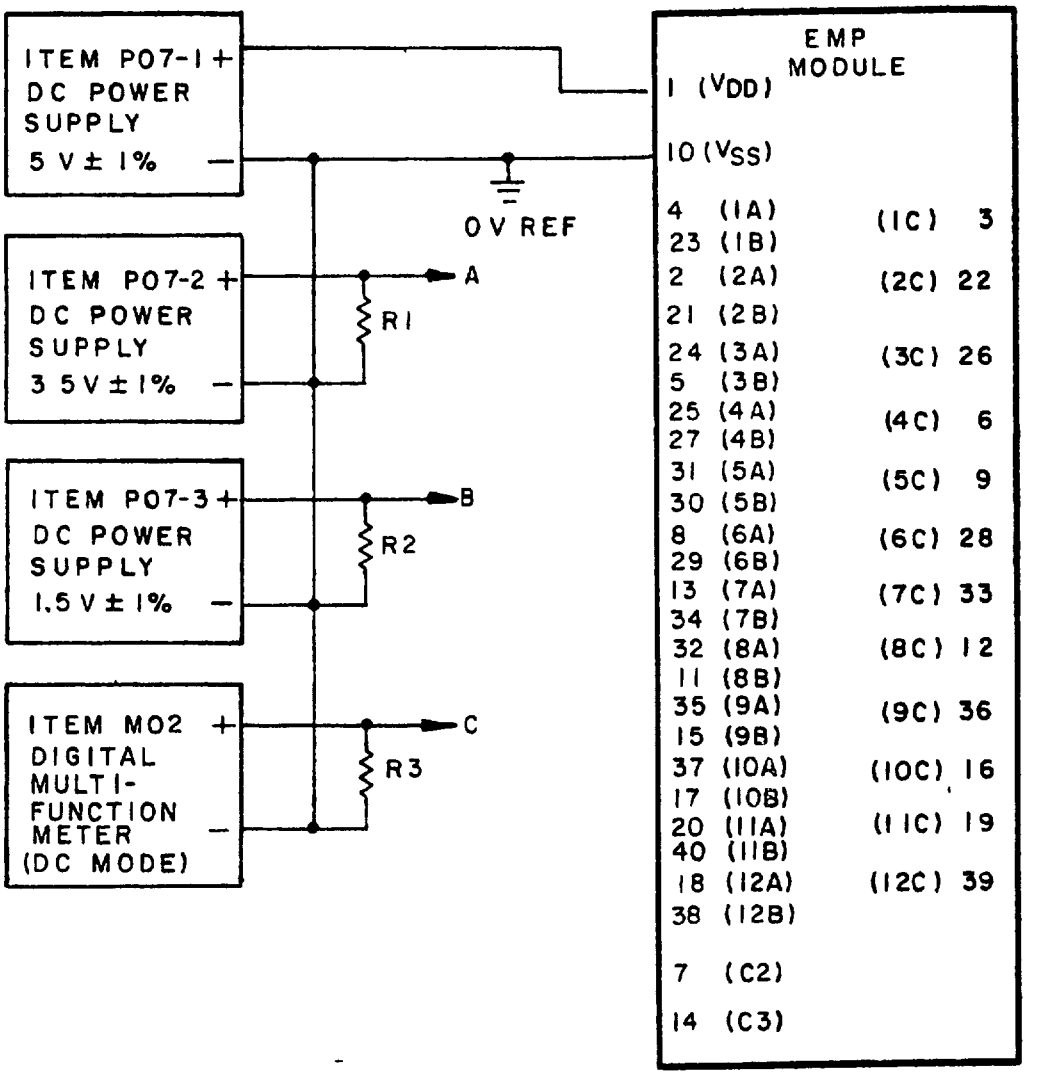
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NOTE: R1 = 1.5 kΩ ±1%, /8 W.

FIGURE 7. Output voltage (high-level, I_{OH} = 0 and low-level, I'_{OL} = 0) test circuit for tables VI and VII.

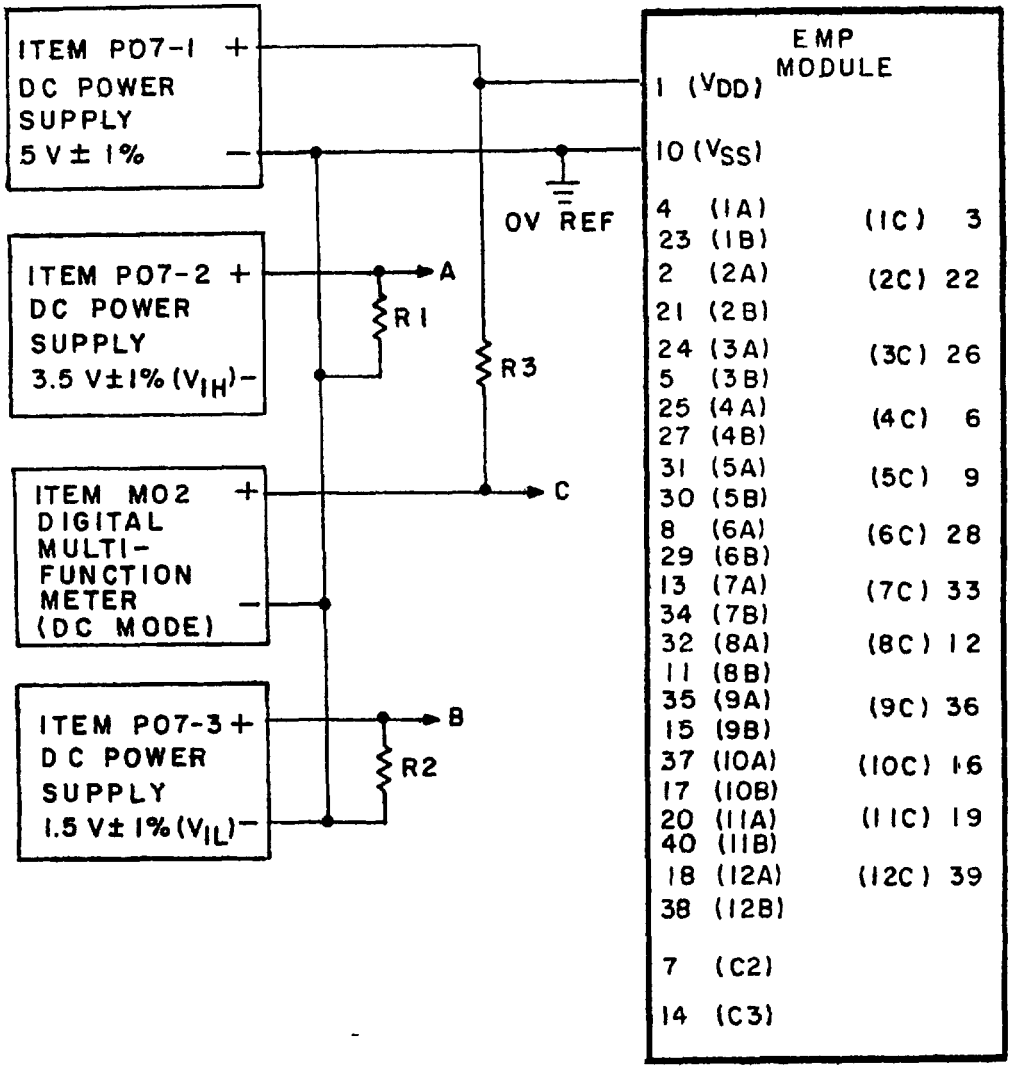
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = At 0° and 25°C - 23.2 kΩ ±1%, 1/8 W.
= At 60°C = 27.4 kΩ ±1%, 1/8 W.
= At -55°C = 18.2 kΩ ±1%, 1/8 W.
= At 85°C = 32.4 kΩ ±1%, 1/8 W.

FIGURE 8. Output voltage (high-level, V_{DD} = 5 V) test circuit for table VIII.

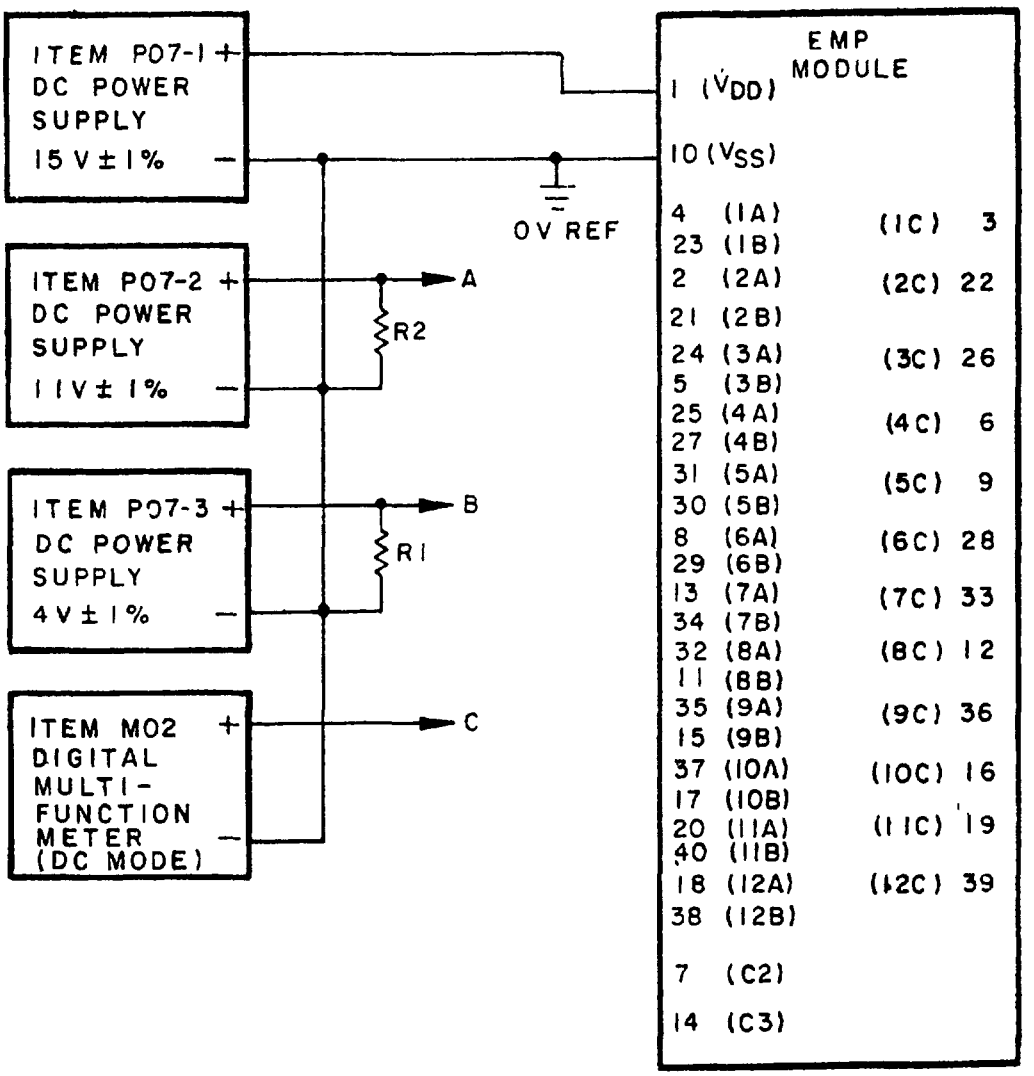
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NOTES:
1. R1 = R2 = 51.1Ω ±5%, 1/2 W.
2. R3 = At 0° and 25°C - 9.09 kΩ ±1%, 1/8 W.
   = At 60°C = 10.5 kΩ ±1%, 1/8 W.
   = At -55°C = 7.15 kΩ ±1%, 1/8 W.
   = At 85°C = 12.7 kΩ ±1%, 1/8 W.

FIGURE 9. Output voltage (low-level, V_{DD} = 5 V) test circuit for table IX.

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NOTES:
1. R1 = 51.1Ω ±5%, 1/2 W.
2. R2 = 576Ω ±5%, 1/4 W.

FIGURE 10. Output voltage (high-level and low-level, V_{DD} = 15 V) test circuit for tables X and XI.

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